

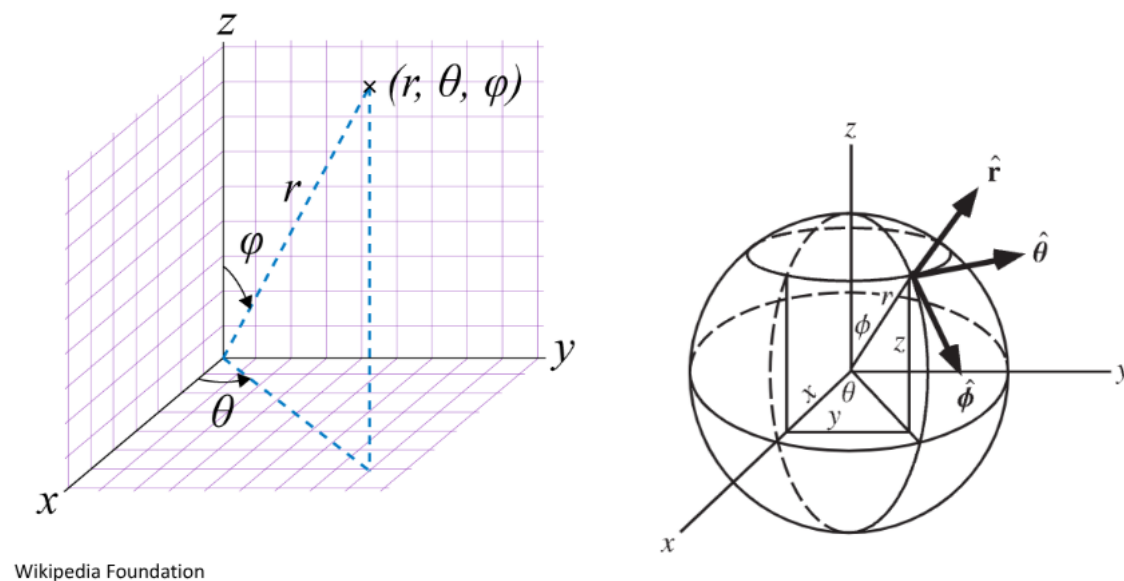
1 Spherical Coordinates

Spherical coordinates, typically denoted as (r, θ, ϕ) , are useful for describing positions on a sphere or spheroid. Spherical coordinates are a powerful three-dimensional coordinate system essential in physics for simplifying problems involving points, spheres, or central forces such as gravity, electrostatics, and quantum mechanics. To better explain physics concepts, this guide will frequently employ spherical coordinates, so it is useful to become familiar with them.

Unlike Cartesian coordinates (x, y, z) , which use three linear distances to locate a point, spherical coordinates define a position using one radial distance r (or ρ) from the origin and two angles: θ and ϕ , which measure inclination from the z -axis and rotation around the xy -plane. While Cartesian coordinates are ideal for linear or box-shaped objects, they make calculations involving circles or spheres very complex. In contrast, a sphere centered at the origin is naturally described in spherical coordinates, reducing complicated three-dimensional problems into manageable and organized calculations.

In terms of Cartesian coordinates:

$$x = r \cos \theta \sin \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \phi$$



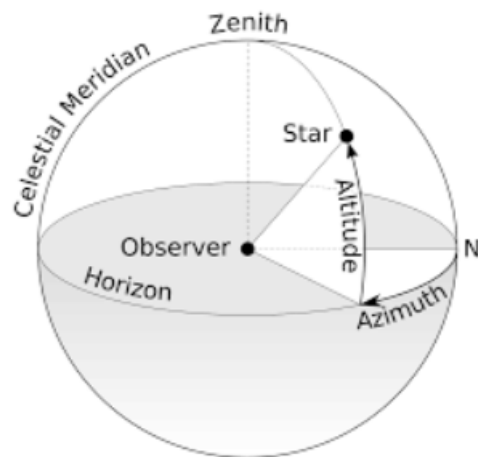
2 Brahe's Astronomical Data

Tycho Brahe originally collected astronomical data of Mars' position in the sky using the Alt-Azimuth Coordinate System, the Ecliptic Coordinate System, and the Equatorial Coordinate System. Later, Johannes Kepler used Brahe's data to develop his three laws of planetary motion.

2.1 Horizontal (Alt-Azimuth) Coordinate System

The Alt-Azimuth Coordinate System, also known as the Horizontal Coordinate System, is a celestial coordinate system that uses the observer's local horizon as the fundamental reference plane. It is one of the simplest and most intuitive systems for locating objects in the sky because it describes positions relative to what an observer directly sees from a specific location on Earth. This system was useful for making direct observations from a specific location on Earth because it described objects relative to the local horizon and allowed Brahe to measure where stars and planets appeared in the sky at a given time.

- **Altitude (a):** 0° at the horizon, 90° at the zenith
- **Azimuth (A):** Compass direction — 0° North, 90° East, etc.



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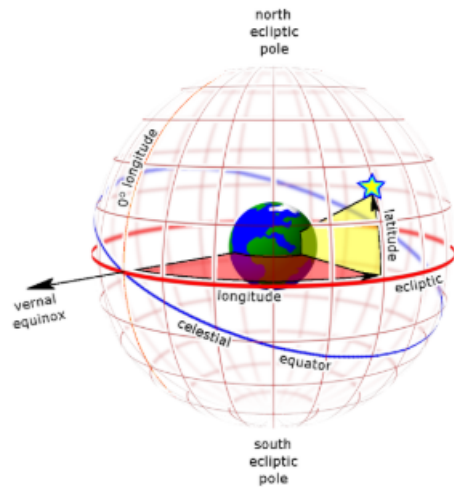
2.2 Ecliptic Coordinate System

The Ecliptic Coordinate System is centered around Earth's orbital plane, known as the ecliptic, around the Sun. It measures two angles: ecliptic longitude and ecliptic latitude.

Ecliptic Longitude (λ) measures the angular distance of a celestial object along the ecliptic plane. It is measured eastward from the vernal equinox, which is the point where the Sun crosses the celestial equator moving northward during the March equinox. Ecliptic longitude ranges from 0° to 360° .

Ecliptic Latitude (β) measures the angular distance of an object above or below the ecliptic plane. Objects located directly on the ecliptic have a latitude of 0° , while objects north of the plane have positive values and objects south of the plane have negative values. Since planets move close to Earth's orbital plane around the Sun, this system made it easier for Brahe to study planetary motion and compare the changing positions of planets.

- **Ecliptic Longitude (λ):** Angle from the vernal equinox
- **Ecliptic Latitude (β):** Angle above or below the ecliptic

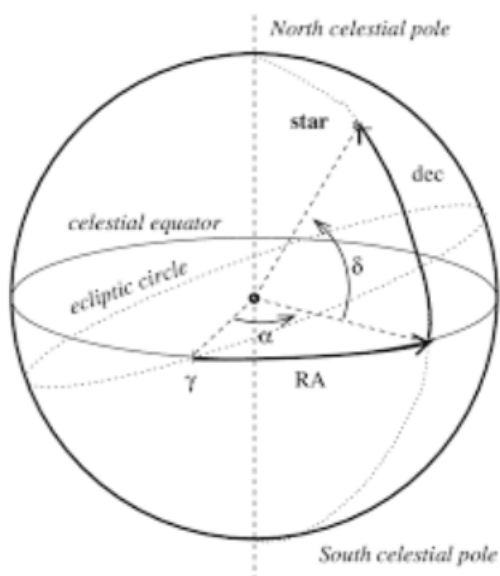


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2.3 Equatorial Coordinate System

The Equatorial Coordinate System is centered around the celestial equator, which is tilted approximately 23.44° relative to the ecliptic plane. This system is independent of the observer's location and remains constant throughout both night and day.

- **Declination (δ):** $+90^\circ$ at the north celestial pole, 0° at the celestial equator, and -90° at the south celestial pole
- **Right Ascension (α):** Measured in hours ($24 \text{ h} = 360^\circ$), starting from the vernal equinox



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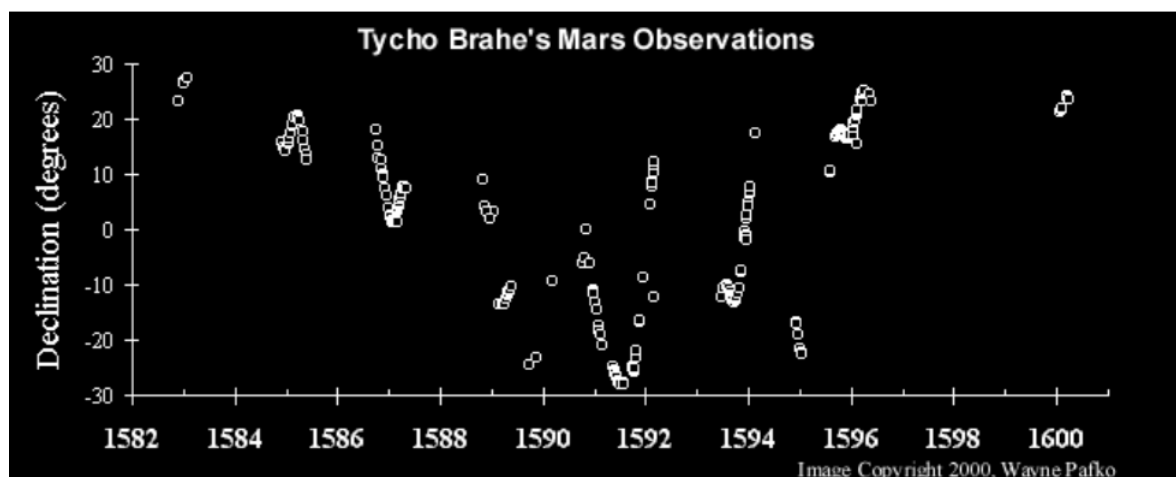
3 Kepler's Laws

Johannes Kepler initially believed that planetary orbits were circular, as had been assumed for centuries. For many of the planets, such as Venus and Jupiter, circular orbit models appeared to work fairly well. However, Mars was an exception due to its higher eccentricity; when Kepler analyzed the motion of Mars using Brahe's observational data, he found that circular models could not accurately match Brahe's precise measurements.

Through mathematical analysis and extensive trial and error of different orbits including offset circles, asymmetric ovals, and symmetric ovals, Kepler eventually determined that planets move in elliptical orbits with the Sun located at one focus. This discovery led to the development of Kepler's First Law of Planetary Motion.

3.1 Why Was Mars Unique?

For many planets, including Venus and Jupiter, circular orbit models appeared to match observations reasonably well. Because the deviations were small, astronomers for centuries believed that planets moved in perfect circles around Earth or the Sun. Mars, however, behaved differently. Its orbit has a noticeably higher eccentricity, meaning its path is more stretched out rather than nearly circular.



Using Tycho Brahe's highly accurate astronomical measurements, Kepler compared the predicted positions of Mars from circular orbit models to Mars's actual observed positions in the sky. When Kepler analyzed Tycho Brahe's extremely precise observations of Mars, he discovered that the planet's predicted position from a circular model consistently disagreed with the actual observed position. Unlike with other planets, the discrepancy for Mars was too large to ignore.

Kepler found a discrepancy of approximately 8 arcminutes between the theoretical predictions and Brahe's data. Although 8 arcminutes is only about one-eighth of the diameter of the full Moon, Kepler considered the error too significant to dismiss because Brahe's instruments were precise enough that such a difference could not simply be

blamed on measurement inaccuracies. Rather than adjusting the data to fit the accepted theory, Kepler trusted the observations and concluded that the model itself was flawed. This small discrepancy ultimately became one of the most important pieces of evidence in astronomy, as it led Kepler to abandon circular orbits and develop the theory of elliptical planetary motion.